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EEE 543 — Past Questions & Materials

by Renance

Course: EEE 543

Institution: Federal University of Petroleum Resources, Effurun

Papers in this pack: 2

Levels: 500

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Paper 1 of 2 — EEE 543 POWER ELECTRONICS 2019,2020

Level 500 • Second Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES. EFFURUN |
COLLEGE OF TECHNOLOGY

Course Code: EEE 543 ;

Course Title; Power Electronics

Time Allowed: 3 Hours

Examination: Second Semester

Session: 2019/2020

Degree: B. Eng,

Date: 21/07/2021

INSTRUCTION; Answer Any five (5) questions of your choice

Se

a (a) With the aid of V-I characteristics of the thyristor explain the following:

(i) Forward blocking region, (ii) Forward Conducting region and (iii) Reverse blocking region.

(b) The thyristor in figure 1 has a holding current of 50 mA and is fired by a gate pulse of length 50 μ s. Show that without resistance R the thyristor will fail to remain on when the firing pulse ends. Find the maximum value of R to ensure firing. Neglect voltage drop across the thyristor.

Figure 1.

2 (a) With the aid of an appropriate diagram(s), explain three methods used in turning on the thyristor.

(b) A d.c supply of 200 V feeds a load having a resistance of 20 Ω and an inductance of 10 H through a thyristor. The latching current of the thyristor is 50 mA. Find the minimum width of the gate pulse.

3 (a) Briefly explain the Resistance triggering circuit of figure 3a

Figure 3a

Figure 3a

230 V, 50 Hz, Resistance R

(b) In the RC triggering circuit of figure 3b, the supply voltage $V_s = 230$ V, $R = 100 \Omega$, $C = 0.01 \mu$ F. Find the range of firing angle α possible with this circuit.

Figure 3b

Figure 3b

Figure 3b

4 (a) With the aid of appropriate circuit diagram, explain the principle of operation of a chopper.

(b)

(b) For a resistive load, develop equations for average load voltage, average load current, RMS output voltage and the output power.

(c) A 250 V separately excited dc motor has an armature resistance of 2.5Ω . When driving a constant torque load at 600 rpm, the armature takes 20 A. The motor is controlled by a chopper with a chopping frequency of 400 Hz. The input voltage to chopper is 250 V. What would be the value of duty ratio, if the speed is reduced from 600 rpm to 400 rpm with load torque maintained constant.

5 (a) Explain the two methods of controlling the duty cycle or output load of a chopper using Z appropriate waveforms.

(b) Use the diagram of figure 5b to explain the effect of commutation in a chopper circuit. c
Figure 5b

6 (a) Draw the circuit diagram and the waveforms of the supply voltage, gate firing pulse, the output voltage and load current of a single-phase half wave controlled converter with commutating diode. Explain its operation.

Paper 2 of 2 — EEE 543 POWER ELECTRONICS 2019,2020

Level 500 • Second Semester • PastQuestion

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Course Code: EEE 543
Course Title: Power Electronics
Time Allowed: 3 Hours

INSTRUCTION: Answer Any five (5) questions of your choice

- 1 (a) With the aid of V-I characteristics of the thyristor explain the following:
(i) Forward blocking region, (ii) Forward Conducting region and (iii) Reverse blocking region.
- (b) The thyristor in figure 1 has a holding current of 50 mA and is fired by a gate pulse of length 50 μ s. Show that without resistance R the thyristor will fail to remain on when the firing pulse ends. Find the maximum value of R to ensure firing. Neglect voltage drop across the thyristor.

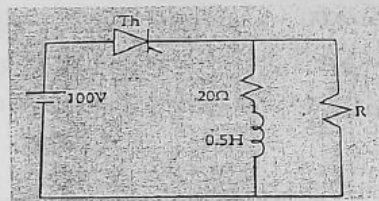


Figure 1.

- 2 (a) With the aid of an appropriate diagram(s), explain three methods used in turning on the thyristor.
- (b) A d.c supply of 200 V feeds a load having a resistance of 20 Ω and an inductance of 10 H through a thyristor. The latching current of the thyristor is 50 mA. Find the minimum width of the gate pulse.
- 3 (a) Briefly explain the Resistance triggering circuit of figure 3a.

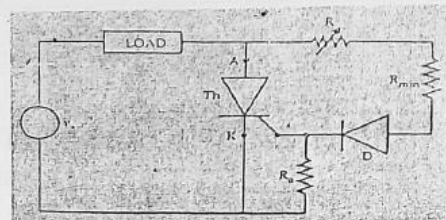


Figure 3a

(b) In the RC triggering circuit of figure 3b, the supply voltage $V_s = 230 \text{ V}_{\text{rms}}$, 50 Hz, Resistance R is varied from $2 \text{ k}\Omega$ to $20 \text{ k}\Omega$. Gate triggering voltage is 2 V . If $C = 0.47 \times 10^{-6} \text{ F}$, find the range of firing angle α possible with this circuit.

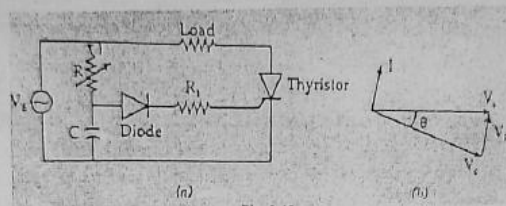


Figure 3b

4 (a) With the aid of appropriate circuit diagram, explain the principle of operation of a chopper.

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(c) A 250 V separately excited dc motor has an armature resistance of 2.5Ω . When driving a constant torque load at 600 rpm , the armature takes 20 A . The motor is controlled by a chopper with a chopping frequency of 400 Hz . The input voltage to chopper is 250 V . What would be the value of duty ratio, if the speed is reduced from 600 rpm to 400 rpm with load torque maintained constant.

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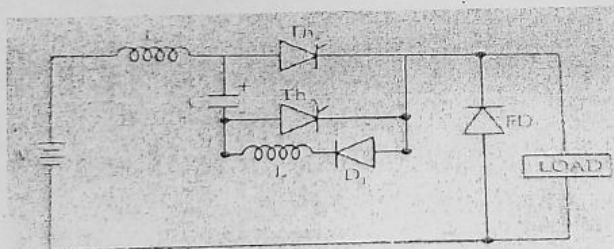


Figure 5b

6 (a) Draw the circuit diagram and the waveforms of the supply voltage, gate firing pulse, the output voltage and load current of a single-phase half wave controlled converter with commutating diode. Explain its operation.